Technology Stack

Date	28 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

The OptiCrop project uses a simple and efficient technology stack. HTML, CSS, and Bootstrap are used for the frontend, Python and Flask for the backend, MySQL for data storage, and Scikit-learn for machine learning. This technology stack ensures a user-friendly, reliable, and scalable application.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (HTML, CSS, Bootstrap)	HTML5, CSS3, Bootstrap 5	Provides responsive and mobile-friendly UI. Bootstrap ensures consistent layout and styling for the crop recommendation input form and results page.
2	Backend / Server-Side (Python, Flask)	Python 3.10+, Flask	Flask is lightweight and ideal for deploying ML models as web APIs. Python offers seamless integration with Scikit-learn and Pandas for data processing.
3	Database / Data Storage (CSV / File-based)	CSV Dataset + Pickle (.pkl)	The agricultural dataset is stored as a CSV file. Trained ML models are serialized using Pickle for efficient reuse during crop prediction without retraining.
4	Cloud / Hosting / Deployment (Local / Flask Dev Server)	Flask Development Server (Local)	The application is deployed locally using the Flask built-in development server, allowing real-time crop recommendations through the browser interface.
5	Version Control & CI/CD (GitHub, Git)	Git + GitHub	Git is used for version control and tracking code changes. GitHub hosts the repository for team collaboration, code review

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn)	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook,	NumPy and Pandas handle data processing. Scikit-learn provides ML algorithms (KNN, Logistic Regression, Random Forest, K-Means). Matplotlib and Seaborn are used for data visualization and analysis.